Beyond the Feature List:

A Decision Paradigm For Evaluating Information Technology Applications And Vendors

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Few Information Technology topics have received the sustained interest visited upon the question of how organizations should embark on the process of selecting software packages. No simple answer exists to this complex question, but a solid understanding of the critical factors that influence the long-term success of a software implementation, combined with a structured approach, can remove both the uncertainty and the risk from this critical decision.

This paper presents a structured decision - making paradigm that takes into account both the nature of the organization, the processes under consideration, the technology environment, and the state of both the packaged solutions and their vendors. It yields sufficient information to allow a selection team to make an informed decision.

We will present a set of decision criteria, discuss the challenges of selecting the correct packaged solution, and finally, outline a structured decision process.

I. DECISION FACTORS

This structured decision paradigm considers nine factors:

The Criteria

- Core vs. Context
- Process
- Coverage
- Direction
- TCO
- Scale
- Complexity
- Technology
- Standards

Core vs. Context

Borrowing the terminology from Geoffrey Moore[i], the first criterion is the strategic significance of the application.

Core activities are those that contribute directly to the organization's differentiation and value creation. Context is everything else.

It is in Core areas that organizations gain strategic advantage and where information systems must conform to business processes, not the reverse.

Mission critical, core activities are the highest priority and deserve the most attention. When assessing the viability of alternative solutions, knowing whether the particular activity is core or context will influence the appropriateness of modifying business practices to fit the strictures of a commercial package.

Process

Solutions must be compared from the perspective of business process, not exclusively from features and functions. One of the most common sources of long-term problems is a reliance on feature-laden technical specifications.

Packages may meet the functional requirements, but fail to provide a solution consistent with the organization's business processes. Conversely, a potential solution may be ruled out because it doesn't match the exhaustive feature list, but would have been and excellent for the business process.

While it is common for solution vendors to claim complete flexibility in configuring their solution to meet existing business process, this is rarely the case. Most often client organizations are urged to modify their business practices to conform to the range of choices that the package offers. This constraint is essential lest the package grow to be unmanageably complex. In practice, this is often not a problem since one of the motivating factors behind the choice of a packaged solution is the desire to re-engineer business processes.

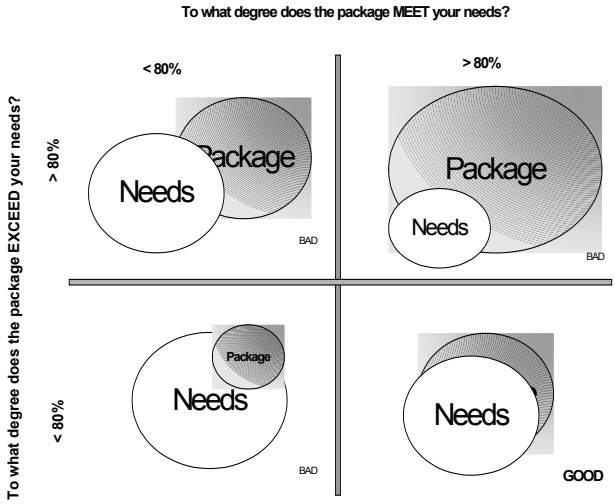
If the application is not strategic, then modifying the organization's process may be more effective than either finding a solution that matches existing practices, or configuring a selected solution. The challenge, however, is not as much one of technology as sociology. Organizations have inherent resistance to change, even in areas that are not core to their value creation and this factor must be considered carefully.

Coverage

Coverage assesses how close the match is between what the business requires and what the packaged solution provides.

The traditional rule of thumb is that packaged solutions must meet a minimum of 80% of the required functionality. Unfortunately, this is an oversimplification.

It is necessary, but not sufficient, to determine if a package covers all the requirements. It is equally important to determine if the package provides capabilities that are outside the requirements. Every feature, every function, every capability represents additional cost and complexity, and those inevitably translate into future costs and complications. Just as it is unwise to consider a package unless it meets 80% of the known needs, similarly, so it is unwise to consider one in which the known requirements represent less than 80% of the package's capabilities.



Buying a solution that has extensive capabilities that you don't need is costly. One way or another, you end up paying for the complexity in terms of training, integration, configuration, maintenance, support, or any of the myriad factors that influence long-term costs. Beyond the pure economic impact, the vendor will be revising, enhancing, and expanding all of those additional features as well. If the package does so much more than you want and need, then it is likely that it is meant to serve a different audience, and your requirements stand a high probability of diverging over time.

Direction

Effective vendor selection must consider not only how well current requirements are covered, but also how the strategic direction of the vendor matches that of the organization. This assessment must consider issues of technology and marketplace. How flexible, maintainable, and extensible will the application be throughout its intended life? What markets is the vendor targeting? New capabilities likely will be intended to meet the needs not of current clients, but of future ones.

When we consider direction we must take into consideration how much control the organization will have over the crafting and addition of desirable, but as yet unknown, new features.

The consideration of direction applies equally to application functions as well as to platform technologies. Both must be considered carefully. Architecture, technology, and integration methods all influence the long-term direction and evolution of the package. They are key indicators of where the package will be, how long it is likely to be enhanced, and consequently, how likely it is that the solution will continue to be a good fit in the future.

Total Cost of Ownership (TCO)

TCO includes not only the cost of acquisition, configuration, and customization, but also the ongoing support, maintenance, and evolution of the application. It is quite common for lifetime costs to dwarf acquisition costs. Perhaps the most important determination in the calculation of TCO is an estimate of the anticipated economic life of the application. TCO must take into account the fixed costs associated with mastering the underlying structure and technology of a packaged solution as well.

Like custom solutions, which require continued availability of development resources to respond to changing requirements, packaged solutions require resources to test, validate, integrate, and support new releases from the vendor. The degree of effort is highly dependent on the nature of the application itself.

Most packaged solutions have both high volatility and a short economic life. They tend to change more often and in dramatic ways as the vendor responds to the demands of competitive pressures, evolving technologies, and the detailed requirements of new customers. Both of these factors can contribute to an increase in operating costs, and should be factored into the decision matrix.

Scale

The larger the scale of the application, the more important it is that it represent core business functions. Conversely, many large, integrated ERP solutions have such a large fixed support cost that it is not reasonable or appropriate to consider implementing a small subset of the complete suite.

Scale becomes an important factor in measuring and ultimately mitigating risk for the project.

Scale also influences rollout strategies. Many an IT project has been undone by staggering implementation costs that were the result of organizational resistance. Segmenting large scale projects into a succession of small phases, selecting the first targets carefully, and building one success upon another can mitigate the risk and reduce the cost.

Complexity

Software has the unfortunate characteristic that the more complex it becomes, the harder it is to extend, modify, or support. Commercial solutions become the victim of their own success: more customers means greater diversity of requirements. The complexity needed to support that diversity shortens the time over which they can meet evolving needs because it becomes too difficult to enhance. It is not uncommon for packaged applications to have a data model with many times the number of tables that the organization requires.

Technology

New customers of packaged solutions demand that those solutions be implemented using the latest and greatest technologies. This is not at all unreasonable, given that new customers are looking for new, advanced solutions. However, technology platforms remain useful for much longer than they are popular. In fact, maturity is a factor that strongly influences reliability.

No organization would choose to acquire a COBOL-based application today, yet large systems in many industries continue to function perfectly well with them. The result is pressure on vendors to migrate to newer technologies faster than the underlying system requires. If they don't, they will become uncompetitive. Thus, it is the requirements of the "next" customer that drives the actions of package vendors, rather than the needs of the user base.

Standards

The existence of established, externally regulated standards influences the economics of vendor selection in a variety of ways.

An external force ensures consistency of a large portion of the requirements. Common examples of such forces are government standards (tax laws, accounting regulations), nature (software to do simulation, for example), broad industry standards (HTML standards allow commercial browsers to exist, ISO standards for mechanical drafting), or powerful industry groups (FIX for financial transfers, EDI, AIA standards for architecture, and others). In such cases there can be some reasonable assurance that the vendor's investments in ongoing development will benefit all users, not just the next ones in line.

Standards may also arise directly from the success of a particular package in areas where organizations are highly likely to see the function as "context," not "core". Office applications (word processing, spreadsheet, email clients) are good examples. Tools and components also fall into that category. Few organizations would think to develop their own

database technology, messaging middleware, or operating system.

In the absence of a strong external force defining consistent content, it is likely that alternative solutions will diverge over time (see "Direction" above); that the feature-function evolution will cease to be a good fit (see "Coverage"); that the TCO will grow as organizations cope with a mismatch between the package and the business requirements. Worst of all, the business users may discover that they cannot ensure that evolving software will match their business process requirements – particularly for functions that are "core."

2. DECISION PROCESS

The vendor selection paradigm we advocate seeks to understand the business needs, and then eliminate candidate solutions that fail to meet those needs. The remaining choices, all considered "acceptable," are then compared from the balanced perspectives of cost and risk. To accomplish this, we use a six-step process:

The Process

1. Assess Organizational Bias
2. Determine Core vs. Context
3. Document scenario-based requirements
4. Review available packages for fit.
5. Develop TCO estimates for all best-fit alternatives
6. Prepare Risk/Mitigation Matrix

Assess Organizational Bias

We start here because all organizations have an innate bias towards specific solutions. That predisposition should be acknowledged, and the reasons for it understood so that they can be addressed directly. The position may be well founded, or based on past experiences that are no long directly application.

If organizational bias is not understood and made explicit, the assessment team can waste substantial time and effort and not provide the information needed to reach a reasoned conclusion

Determine Core vs. Context

It is quite common for organizations to lack clarity in this area, or to confuse traditional business processes with strategically significant ones. Since the choice of a vendor will almost certainly result in the transition from current processes to the ones the package supports - at least at a detail level – it is extremely important to gain insight and concurrence on the strategic significance of the business processes the application will support.

There is ample evidence to lead us to one type of solution for context activities and an alternative solution for core, differentiating activities.

Document Requirements

We strongly advocate the use of scenario-based requirements, supplemented by functional requirements where appropriate. This approach (commonly referred to as "Use Cases") ensures that the chosen solution will meet the business objectives. Requirements that are limited to a Functional Specification miss all of the dynamic aspects of system usage. Worst of all are feature/function comparative checklists.

At this stage of the process, it is useful to approach the process as if you were going to build the whole thing, documenting what you want and need in a way that would let a development team understand what they are to build. Avoid simply prioritizing features and functions, and stick to scenario-based requirements supplemented by high-level, non-functional needs (such as security, availability, access).

Review Packages for Fit

This is where the bulk of the work gets done. Concentrate on both coverage and direction since you need to be sure that the package will meet not just your current needs, but your future needs as well.

When assessing coverage, consider that the 80% rule should apply in both directions – both

what is in, and what is out. A package is a bad fit if either

- It doesn't meet 80% of your needs, or
- Your needs represent less than 80% of what the package does.

Finally, fit should be measured in the non-functional domains as well. The underlying technology, support practices, frequency of release and upgrades, integration tools and capabilities, are among the characteristics worthy of assessment.

TCO Estimates

The economic phase of the determination requires a solid understanding of the total cost of ownership, not simply the cost to acquire, configure, and implement.

TCO calculation needs to take into account the startup costs, as well as the economic life and volatility of the package. How far through its life cycle is it? How often are major and minor releases issued? Is it mandatory to test, integrate, install, and support each of these releases?

It is common for a packaged solution to have very large fixed support costs, many of which are not discussed during the sales cycle, and not at all visible until far into the implementation phase.

Packages that involve substantial configuration, for example, may require elaborate and costly efforts to maintain control over the configuration as it evolves overtime.

Risk and Mitigation

All packages involve risk, some less and some more. Each of the "acceptable" alternatives should be the object of a detailed risk and mitigation review. The combination of increased complexity and technology migration results in an economic life for many packaged solutions that does not meet the goals or desires of the organization. The key is to select a solution that will remain highly productive ten, fifteen, or twenty years after their implementation.

Organizations face this decision process every time they turn to information technology to gain efficiency, improve productivity, or improve their strategic advantage.

4. ASSESS THE VENDOR

Last, but not least, the vendor selection process must include a detailed assessment of the vendor itself, as separate and distinct from the product offering. When the solution is an important business component, the result is a long relationship, a partnership more than merely a buyer/seller transaction.

Vendor assessment will include interviews with existing customers, literature review, analyst reviews (if the vendor is a public company), and discussions with vendor staff in a variety of professional capacities.

References:

[i] **Living on the Fault Line: Managing for Shareholder Value in Any Economy**, Moore, Geoffrey, HarperCollins, 2002